

A Σ_2^0 reduction from $\mathcal{P}_\infty$ to admissible $SB(X)$

Run `fa_banach.001`; source arXiv:1912.03994

Status

candidate full solution to Question 2 of arXiv:1912.03994.

Cuth–Dolezal–Doucha–Kurka ask whether, for every isometrically universal separable Banach space X and every admissible topology τ on $SB(X)$, there is a Σ_2^0 -measurable map

$$\Phi : \mathcal{P}_\infty \rightarrow (SB(X), \tau)$$

such that X_μ is linearly isometric to $\Phi(\mu)$ for every $\mu \in \mathcal{P}_\infty$. This packet proves that the answer is yes.

The argument does *not* answer Question 1, which asks for a continuous map $\mathcal{P}_\infty \rightarrow \mathcal{B}$. Instead it proves the sufficient condition for Question 2 highlighted in the source paper, using the continuous dual-ball coding from the later paper arXiv:2204.06834.

Source Questions

The source questions appear on pages 20–21 of arXiv:1912.03994.

for every $\bigwedge_{k=1}^\infty \exists e_k \in c_{00}$ and every $\mu \in \mathcal{B}$. □

To summarize, we obtained an optimal reduction from $\mathcal{B}$ to $SB_\infty(X)$ and from $SB_\infty(X)$ to $\mathcal{P}_\infty$. However, our reduction from $\mathcal{P}_\infty$ to $\mathcal{B}$ seems not to be optimal, so one is tempted to ask the following.

Question 1. Does there exist a continuous mapping $\Phi : \mathcal{P}_\infty \rightarrow \mathcal{B}$ such that for every $\mu \in \mathcal{P}_\infty$ we have $X_\mu \equiv X_{\Phi(\mu)}$?

POLISH SPACES OF BANACH SPACES

21

Note that a positive answer to Question 1 would imply a positive answer to Question 2 and that a sufficient condition for a positive solution of Question 2 is provided by Proposition 2.15.

Question 2. Let X be an isometrically universal separable Banach space and let τ be an admissible topology on $SB(X)$. Does there exist a Σ_2^0 -measurable mapping $\Phi : \mathcal{P}_\infty \rightarrow (SB(X), \tau)$ such that for every $\mu \in \mathcal{P}_\infty$ we have $X_\mu \equiv \Phi(\mu)$?

Question 1 asks for a continuous reduction $\mathcal{P}_\infty \rightarrow \mathcal{B}$. Question 2 asks for the corresponding Σ_2^0 -measurable reduction from $\mathcal{P}_\infty$ to $SB(X)$ with an admissible topology.

Idea of the Proof

The source paper proves that Question 2 would follow if every pseudonorm code could be realized by continuous vectors $\chi_k(\mu)$ in one separable Banach space:

$$\left\| \sum_{k=1}^n a_k \chi_k(\mu) \right\| = \mu \left(\sum_{k=1}^n a_k e_k \right).$$

The authors prove this for actual norms after normalizing all coordinates, but state that they do not know it for pseudonorms.

The missing ingredient is to work on the dual side. The sequel arXiv:2204.06834 constructs, continuously in μ , the weak-star compact dual unit ball $\Omega(\mu)$ as a compact convex subset of the Hilbert cube. For compact convex subsets of a Hilbert space, projecting a fixed dense sequence onto the moving compact convex set gives continuous dense selections. Evaluating the corresponding dense sequence of dual functionals on the coordinates produces the desired vectors $\chi_k(\mu)$ in a separable subspace of ℓ_∞ .

A projection parametrization lemma

Let $Q = \prod_{m=1}^\infty [-1, 1]$, and embed Q linearly and homeomorphically into the Hilbert space ℓ_2 by

$$J((a_m)_{m=1}^\infty) = (2^{-m} a_m)_{m=1}^\infty.$$

Write $Q_H = J(Q)$.

Lemma 1. *There are continuous maps*

$$p_r : \mathcal{K}_c(Q_H) \rightarrow Q_H, \quad r \in \mathbb{N},$$

where $\mathcal{K}_c(Q_H)$ denotes the hyperspace of nonempty compact convex subsets of Q_H , such that $\{p_r(F) : r \in \mathbb{N}\}$ is dense in F for every $F \in \mathcal{K}_c(Q_H)$.

Proof. Fix a countable dense set (d_r) in Q_H . For $F \in \mathcal{K}_c(Q_H)$, let $p_r(F)$ be the Hilbert-space metric projection of d_r onto F . This projection exists and is unique because F is nonempty, compact, and convex.

We first check continuity. Suppose $F_n \rightarrow F$ in the Hausdorff metric and $z_n \rightarrow z$ in Q_H . Put $y_n = P_{F_n}(z_n)$. Passing to a subsequence if necessary, compactness gives $y_n \rightarrow y \in F$. For every $x \in F$, choose $x_n \in F_n$ with $x_n \rightarrow x$. Since y_n is the metric projection,

$$\|z_n - y_n\| \leq \|z_n - x_n\|.$$

Passing to the limit gives $\|z - y\| \leq \|z - x\|$ for every $x \in F$. Thus $y = P_F(z)$, by uniqueness of the projection. Hence $(F, z) \mapsto P_F(z)$ is continuous on $\mathcal{K}_c(Q_H) \times Q_H$. In particular, each p_r is continuous.

Finally let $x \in F$ and $\varepsilon > 0$. Pick r with $\|d_r - x\| < \varepsilon$. Since $P_F(d_r)$ is at least as close to d_r as x is,

$$\|p_r(F) - d_r\| \leq \|x - d_r\| < \varepsilon.$$

Therefore $\|p_r(F) - x\| < 2\varepsilon$, proving density. □

Continuous realization of all pseudonorms

Let V be the countable rational vector space with basis $(e_k)_{k \in \mathbb{N}}$. Fix an enumeration $(v_m)_{m \in \mathbb{N}}$ of V such that every e_k occurs, and let $m(k)$ be an index with $v_{m(k)} = e_k$. Let $g : [0, \infty) \rightarrow [1, \infty)$ be $g(t) = 1$ for $t \leq 1$ and $g(t) = t$ for $t > 1$.

We use the following result from arXiv:2204.06834, the lemma labelled Ω in the subsection “Dual unit balls and the Szlenk derivative.” For $\mu \in \mathcal{P}$, define $x_m^\mu = v_m/g(\mu(v_m)) \in B_{X_\mu}$. The set $\Omega(\mu) \subseteq Q$ of all sequences

$$(x^*(x_m^\mu))_{m=1}^\infty, \quad x^* \in B_{X_\mu^*},$$

is a weak-star compact convex subset of Q , and the map

$$\Omega : \mathcal{P} \rightarrow \mathcal{K}(Q)$$

is continuous. Moreover $\Omega(\mu)$ is exactly the dual unit ball of X_μ written in the coordinates x_m^μ .

Theorem 2. *There are a separable Banach space U and continuous maps*

$$\chi_k : \mathcal{P} \rightarrow U, \quad k \in \mathbb{N},$$

such that for every $\mu \in \mathcal{P}$ and every finite scalar family $(a_k)_{k=1}^n$,

$$\left\| \sum_{k=1}^n a_k \chi_k(\mu) \right\|_U = \mu \left(\sum_{k=1}^n a_k e_k \right).$$

Proof. For $\mu \in \mathcal{P}$, put $F_\mu = J(\Omega(\mu)) \in \mathcal{K}_c(Q_H)$. Let p_r be the maps from Lemma 1, and put

$$q_r(\mu) = J^{-1}(p_r(F_\mu)) \in \Omega(\mu).$$

For each $k \in \mathbb{N}$, define a bounded scalar sequence

$$\chi_k(\mu) = (g(\mu(e_k)) q_r(\mu)_{m(k)})_{r=1}^\infty \in \ell_\infty.$$

The map $\mu \mapsto \chi_k(\mu)$ is continuous as a map into ℓ_∞ . Indeed, $\mu \mapsto F_\mu$ is continuous; the projection map $(F, z) \mapsto P_F(z)$ is continuous on the compact space $\mathcal{K}_c(Q_H) \times Q_H$, hence uniformly continuous there; and the fixed coordinate $J^{-1}(\cdot)_{m(k)}$ is continuous on Q_H . The scalar factor $g(\mu(e_k))$ is also continuous.

Let U be the closed linear span in ℓ_∞ of

$$\bigcup_{k \in \mathbb{N}} \chi_k(\mathcal{P}).$$

Since $\mathcal{P}$ is separable and every χ_k is continuous into the metric space ℓ_∞ , each $\chi_k(\mathcal{P})$ is separable. Therefore U is separable.

It remains to compute the norm. Fix $\mu \in \mathcal{P}$ and $u = \sum_{k=1}^n a_k e_k$. Since $\{q_r(\mu) : r \in \mathbb{N}\}$ is dense in $\Omega(\mu)$,

$$\begin{aligned} \left\| \sum_{k=1}^n a_k \chi_k(\mu) \right\|_{\ell_\infty} &= \sup_{r \in \mathbb{N}} \left| \sum_{k=1}^n a_k g(\mu(e_k)) q_r(\mu)_{m(k)} \right| \\ &= \sup_{b \in \Omega(\mu)} \left| \sum_{k=1}^n a_k g(\mu(e_k)) b_{m(k)} \right|. \end{aligned}$$

If $b \in \Omega(\mu)$, then $b_{m(k)} = x^*(e_k/g(\mu(e_k)))$ for some $x^* \in B_{X_\mu^*}$. Hence the last supremum is

$$\sup_{x^* \in B_{X_\mu^*}} |x^*(u)| = \|u\|_{X_\mu} = \mu(u),$$

by the Hahn–Banach theorem. This proves the theorem. $\square$

Answer to Question 2

Corollary 3. *Let X be an isometrically universal separable Banach space and let τ be an admissible topology on $SB(X)$. There exists a Σ_2^0 -measurable map*

$$\Phi : \mathcal{P}_\infty \rightarrow (SB(X), \tau)$$

such that X_μ is linearly isometric to $\Phi(\mu)$ for every $\mu \in \mathcal{P}_\infty$.

Proof. By Theorem 2, choose U and $\chi_k : \mathcal{P} \rightarrow U$. Since X is isometrically universal, regard U as a subspace of X . Define

$$\Phi(\mu) = \overline{\text{span}}\{\chi_k(\mu) : k \in \mathbb{N}\} \subseteq X.$$

The norm identity from Theorem 2 says that the linear map $e_k \mapsto \chi_k(\mu)$ extends to an onto linear isometry from X_μ to $\Phi(\mu)$.

It remains only to record the descriptive complexity. The source paper, Lemma 2.10, proves that whenever the realizing maps χ_k are Σ_n^0 -measurable, the map $\mu \mapsto \overline{\text{span}}\{\chi_k(\mu)\}$ into any admissible $SB(X)$ topology is Σ_{n+1}^0 -measurable. Here the maps χ_k are continuous, i.e. Σ_1^0 -measurable, so Φ is Σ_2^0 -measurable. Restricting the domain to $\mathcal{P}_\infty$ gives the desired map. $\square$

Verification and limitations

The proof is non-computational. The key continuity point is the uniform continuity of Hilbert-space projection on the compact set $\mathcal{K}_c(Q_H) \times Q_H$; this gives continuity of the coordinate maps into ℓ_∞ , not merely pointwise continuity of each scalar coordinate.

This answers Question 2 as stated. It does not answer Question 1: the construction realizes X_μ as a closed subspace of a fixed universal Banach space, but it does not produce a continuous norm code $\mathcal{P}_\infty \rightarrow \mathcal{B}$ with an independent coordinate sequence.

Novelty check

Local index searches for arXiv:1912.03994, $\mathcal{P}_\infty$, $\mathcal{B}$, Σ_2^0 , admissible $SB(X)$, and the source questions found only the prior fixed-representative partial packet for Question 1. Web searches on 2026-06-16 for the exact question and author/title phrases found arXiv:1912.03994 and the sequel arXiv:2204.06834, but no statement that Question 2 had been answered. The present proof uses the continuous dual-ball lemma from arXiv:2204.06834 as an input and adds the projection-parametrization argument.

References

References

- [1] M. Cuth, M. Dolezal, M. Doucha, and O. Kurka, *Polish spaces of Banach spaces*, arXiv:1912.03994.
- [2] M. Cuth, M. Dolezal, M. Doucha, and O. Kurka, *Polish spaces of Banach spaces. Complexity of isometry and isomorphism classes*, arXiv:2204.06834.